

## **SWC/Precipitation Finalized Anomaly Detection**

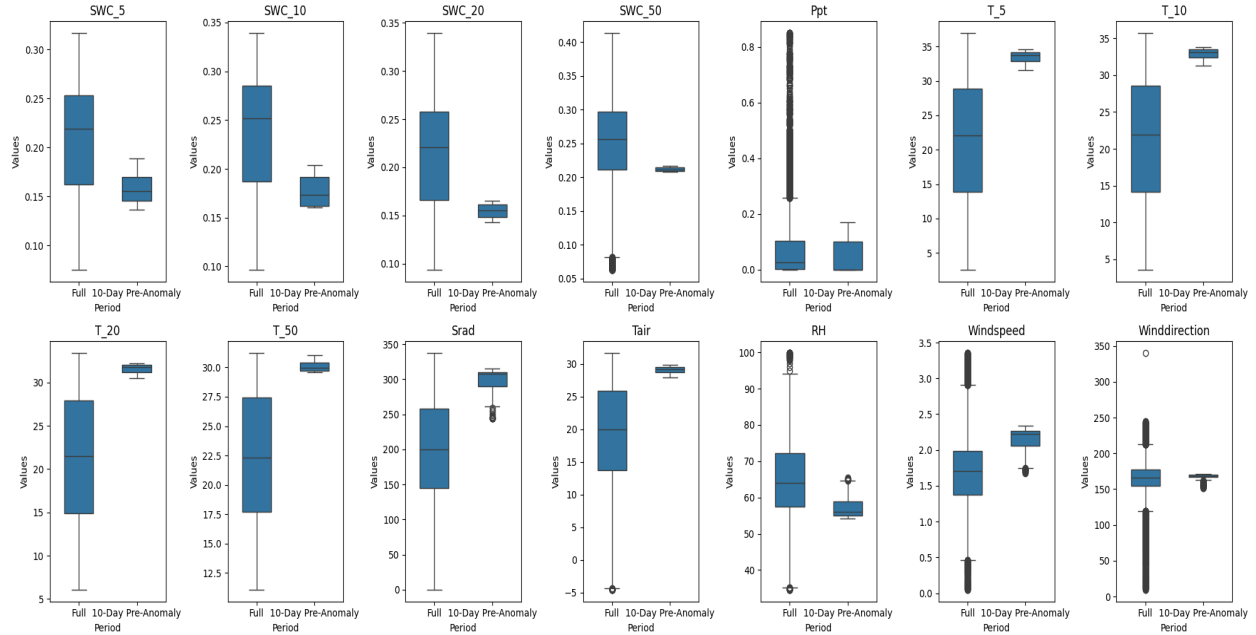
### **(SWC\_Precipitation\_Anomaly\_Detection\_Final.ipynb)**

Autoencoders were trained on the 'Ppt' and 'SWC\_5' features for each station in order to detect the anomalies in each feature for each station. Additionally, the autoencoders were trained with varying window sizes (1, 2, 3, 7, and 10-day) in order to analyze how window sizing affected which timestamps were flagged as anomalies. In total, there were 30 (6 stations and 5 window sizes) autoencoder models trained for each feature. The anomalies corresponding to each feature and station are stored in the corresponding {feature}\_Station{n} folder in the 'Anomaly Dates' folder. For example, the 'Ppt\_Station1' folder contains the anomalies detected in the Station 1 precipitation data for each of the 5 window sizes. Each one of these folders contains 5 files for each of the 5 window sizes that were tested.

The 'Anomaly Dates' folder also contains the 'Ppt\_High\_Frequency\_Anomalies' and the 'SWC\_5\_High\_Frequency\_Anomalies' folders. Each one of these folders contains 7 files. There are 6 files titled '{feature}\_Station\_{StationNumber}\_High\_Frequency\_Anomalies.csv'. Each one of these files contains the anomalies that were present across all 5 window sizes for the specified feature and station. For example, 'Ppt\_Station\_1\_High\_Frequency\_Anomalies.csv' contains the anomalies in the Station 1 precipitation data that appeared in all 5 window sizes (1, 2, 3, 7, and 10-day). The seventh file in each of these folders is titled '{feature}\_Common\_Dates\_Across\_Stations.csv'. The file contains the anomaly dates that appeared multiple times in the 6 "High\_Frequency\_Anomalies" files described previously across all stations. Additionally, the file contains the count for how many stations the specific anomaly date appeared in (the maximum value for this count is 6 as there are 6 stations).

For the anomaly dates that appeared the greatest number of times in the '{feature}\_Common\_Dates\_Across\_Stations.csv' file, boxplots were generated for each feature to compare the overall feature distribution to the feature distribution in the 10 days before the anomaly. For each anomaly date, the feature distributions are generated for each station that the anomaly date appeared in. For example, if there is an anomaly date that appeared in the "High\_Frequency\_Anomalies" files for Stations 1 and 2, two separate feature distribution plots are generated to visualize the disparity between the overall and current distribution. An example of the distribution plot is shown below.

2016-08-16 Feature Distribution at Station 4

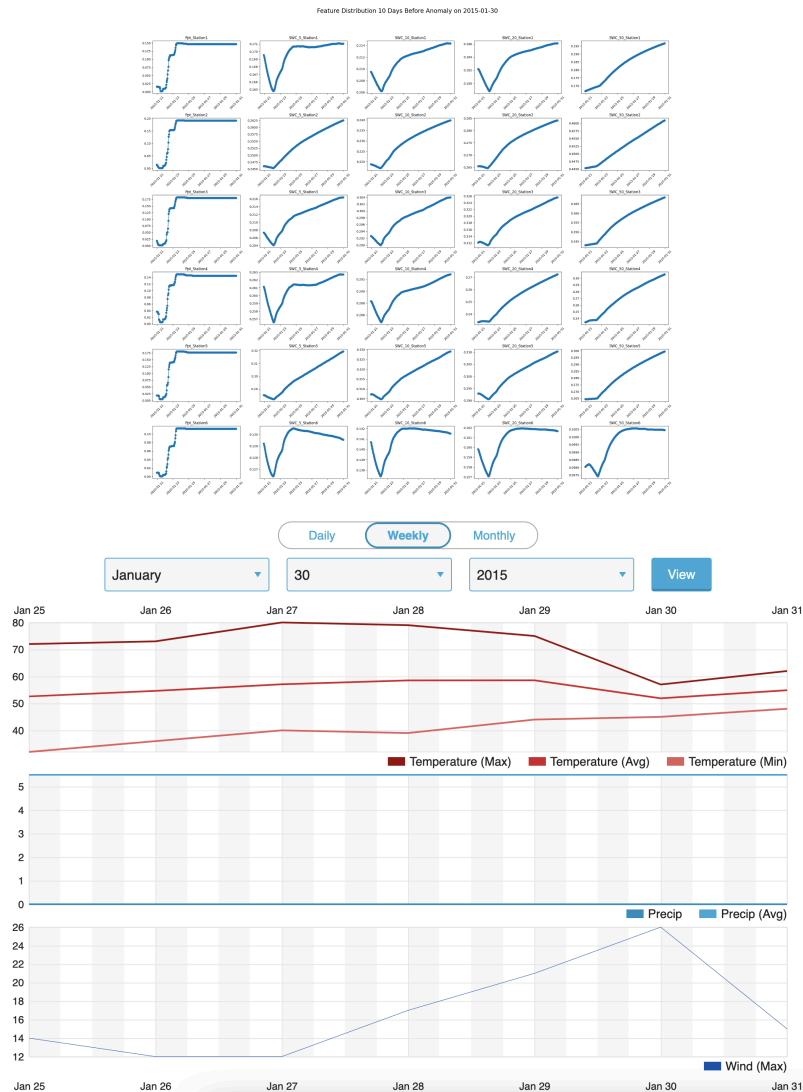


The feature distribution plots can provide more insight into why the specific date was flagged as an anomaly. For example, August 16, 2016, was likely flagged as an anomaly due to the outlier precipitation levels present in the Station 4 data. Additionally, the soil temperature is markedly higher in the 10 days before the anomaly date compared to the overall distribution. This form of analysis can help explicitly denote the contributing factors to each anomaly.

The SWC\_5 anomaly dates that appeared most frequently across all 6 high-frequency anomaly lists were August 15, 2016, August 16, 2016, and September 10, 2018.

## Multivariate Anomaly Detection (Multivariate\_Autoencoder\_Anomaly\_Detection.ipynb)

The multivariate LSTM autoencoder detected three anomalies that occurred on January 30, 2015, February 6, 2015, and February 13, 2015. The anomaly on January 30, 2015, was primarily caused by a spike in precipitation from approximately 0 to 0.15 - 0.20 mm across all six stations on January 22, 2015. This event is reflected in every other feature as the SWC\_5, SWC\_10, SWC\_20, and SWC\_50 readings all dramatically increased on January 22, 2015.

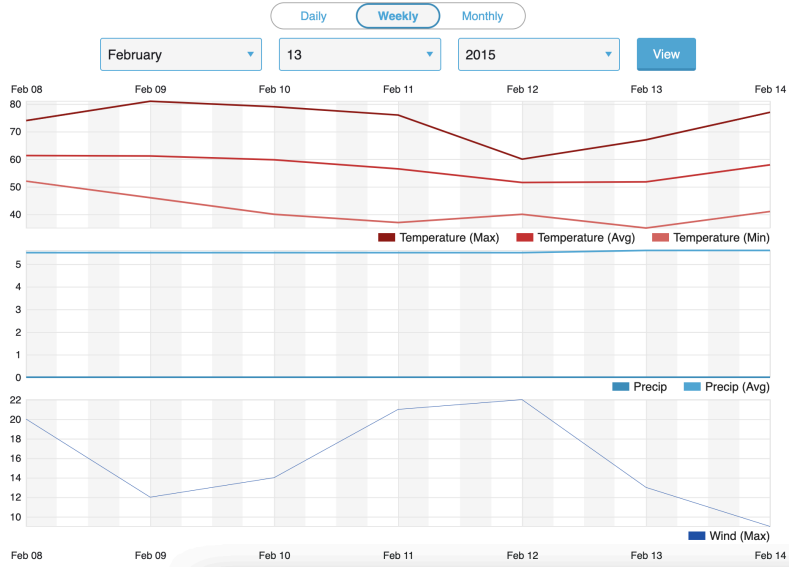
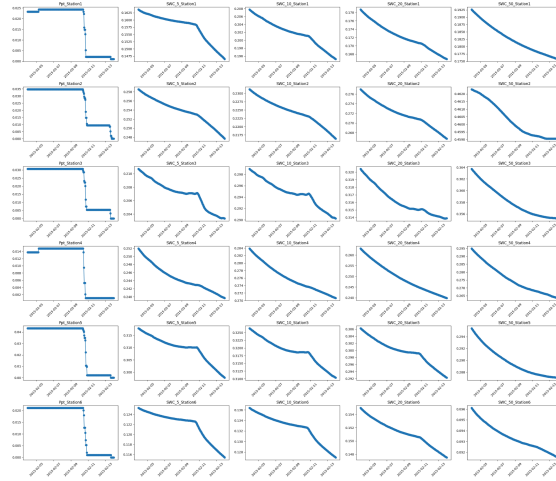


The anomaly on February 6, 2015, was primarily caused by a dropoff in precipitation values from 0.18 - 0.20 mm to 0.02 - 0.05 mm around February 1, 2015. This is likely a reversion to the mean after the high rainfall that occurred a week earlier as detailed above. This event is reflected in every other feature as the SWC\_5, SWC\_10, SWC\_20, and SWC\_50 readings all dramatically decreased around February 1, 2015.



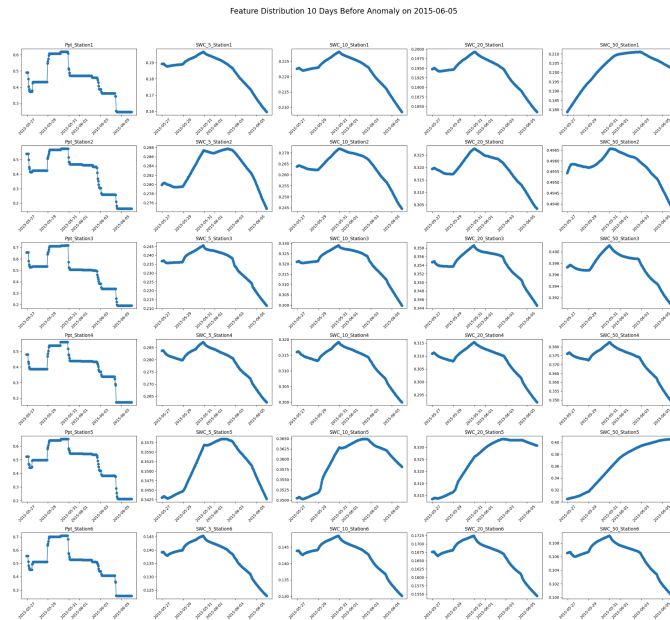
The anomaly on February 13, 2015, was primarily caused by a dropoff in precipitation values from 0.01 - 0.04 mm to 0 mm around February 11, 2015. This dropoff is likely a further reflection of the decrease in rainfall compared to the extreme precipitation in late January 2015. This event is reflected in most other features as most reading dramatically decreased around February 11, 2015, but some stations (especially Station 4) exhibit no noticeable discrepancy between the rates of decline before and after February 11, 2015.

Feature Distribution 10 Days Before Anomaly on 2015-02-13



## Attention-Based LSTM Autoencoder Anomaly Detection: (Multivariate\_Attention\_Autoencoder\_Anomaly\_Detection.ipynb)

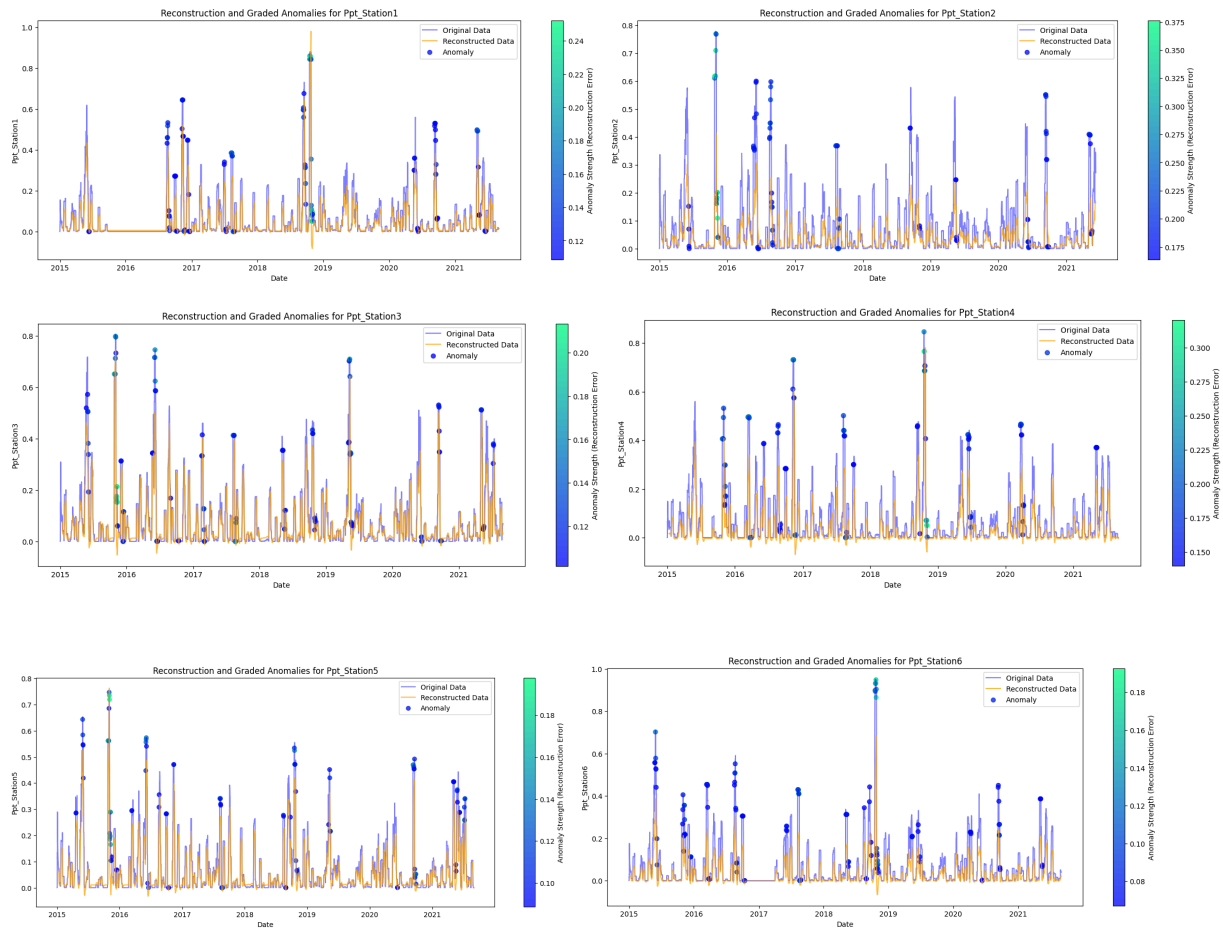
Incorporating attention mechanisms into the existing multivariate LSTM autoencoder generated more anomaly candidates. 19 anomalies were identified by the attention-based autoencoder as opposed to just 3 for the regular multivariate autoencoder. Each of the anomalies and their feature distribution visualizations can be seen in the Multivariate\_Attention\_Autoencoder\_Anomaly\_Detection.ipynb notebook. One of the strongest anomalies occurred on June 5, 2015. The feature distribution plot is shown below.



The anomaly on June 5, 2015, was primarily caused by a dropoff in precipitation values from 0.6 mm to 0.3 mm around May 30, 2015. This time period was characterized by high rainfall, and although the amount of rainfall exceeded normal conditions throughout the period, the rainfall at the end of the 10-day period is more than 50% less than the beginning of the period. This event is reflected in every other feature as the SWC\_5, SWC\_10, SWC\_20, and SWC\_50 readings all dramatically decreased around May 30, 2015.

## Precipitation Anomaly Detection: (Ppt\_AE\_SWC\_LSTM\_Anomaly\_Detection.ipynb)

The strongest precipitation anomalies were found in two distinct periods (October 25, 2015 - November 12, 2015 and October 17, 2018 - November 3, 2018). Each period is characterized by a dramatic increase in rainfall and a subsequent dropoff. The LSTM autoencoder trained on the precipitation time series data flagged almost every day across each of these periods in both the spike and dropoff in precipitation levels.



## LSTM SWC Anomaly Detection: (Ppt\_AE\_SWC\_LSTM\_Anomaly\_Detection.ipynb)

An LSTM model was trained to predict soil water content levels across stations and detect anomalies based on the difference between the predictions and actual data. The strongest anomalies were found in a period from June 2017 - August 2017 as well as from April 2021 - June 2021. Most of the detected anomalies were abnormal increases in SWC levels as opposed to extreme dropoffs.

